

ORACLE

Sovereign Cloud in a Geopolitical Era

and how Oracle is already ahead as a EU Sovereign Hyperscaler

Kim Tholstorf

Master Principal Account Cloud Engineer

kim.tholstorf@oracle.com



Kim Tholstorf

Master Principal Account Cloud Engineer



Experience

Oracle: 7 months
VMware: 11 years
DDIS: 4 years



Expertise

Defence & Intelligence
Industry
High Performance
Computing
Cloud-Native Architecture



Location

Denmark







DK EU



DK EU US  GB AU BR



Business Strategy Customer & Market Regulatory & Legal Security & Risk Cost/Value Politics

EU Compliance and Security Frameworks Complementing GDPR

EU Cloud Code of Conduct (CoC)

Common framework for GDPR compliance across IaaS, PaaS and SaaS (and more)

CISPE Code of Conduct (CoC)

Sector-specific GDPR framework for IaaS

GAIA-X

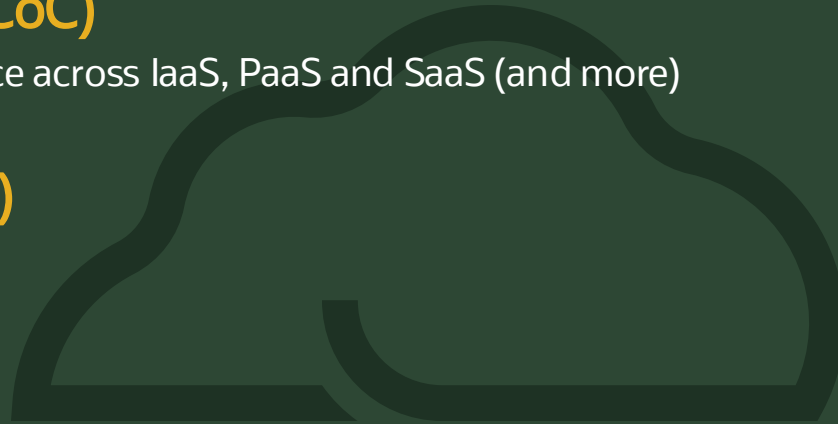
Focus on openness and interoperability / federation (GXFS) for data infrastructure

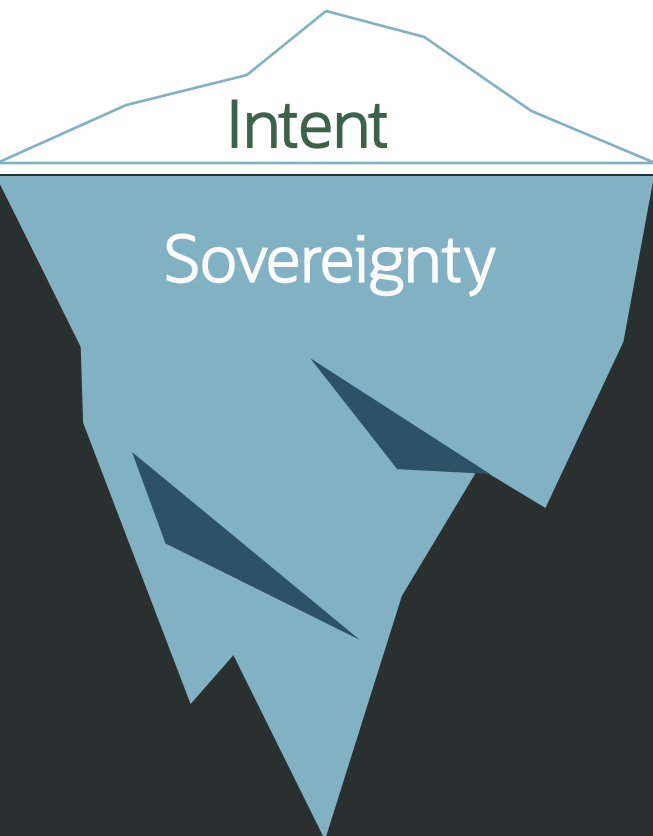
NIS2

Sets requirements for cybersecurity / supply chain / incident reporting in the critical sectors (cloud services, energy, healthcare, etc.)

DORA

Regulates the financial sector and focuses on digital resilience. Contractual rights, exit strategy, audits and EU authorities' direct supervision of cloud providers / hyperscalers





**DATA
SOVEREIGNTY**

**OPERATIONAL
SOVEREIGNTY**

**TECHNOLOGICAL
SOVEREIGNTY**

ORGINAZATIONAL CONTROLS	TECHNICAL CONTROLS	CONTRACTIONAL CONTROLS
------------------------------------	-------------------------------	-----------------------------------

**FULL SUITE OF CLOUD
SECURITY SERVICES**

**SOVEREIGN KEY MGT WITH
EU PARTNER THALES**

DATA CENTER AUDITS

2 EU REGIONS

EU ONLY PROCESSING

**NO DATA TRANSFER
OUTSIDE THE EU**

**INDEPENDENT EU CLOUD
OPERATIONS TEAM**

**INDEPENDENT EU CLOUD
SUPPORT TEAM**

**INDEPENDENT EU DATA
CENTER TEAM**

**EU RESIDENT EMPLOYEES
ONLY**

**ALWAYS ENCRYPTED & HOLD
YOUR OWN KEY**

NO EGRESS FEES

EU ASSET OWNERSHIP

**DATA ACCESS REQUESTS
HANDLING PROCEDURE**

**TECHNOLOGY AVAILABLE
ON-PREMISE**

**ISOLATET & DEDICATED
REALM (CONTROLPLANE)**

RESILIENCE WITHIN THE EU

CHANGE MANAGEMENT

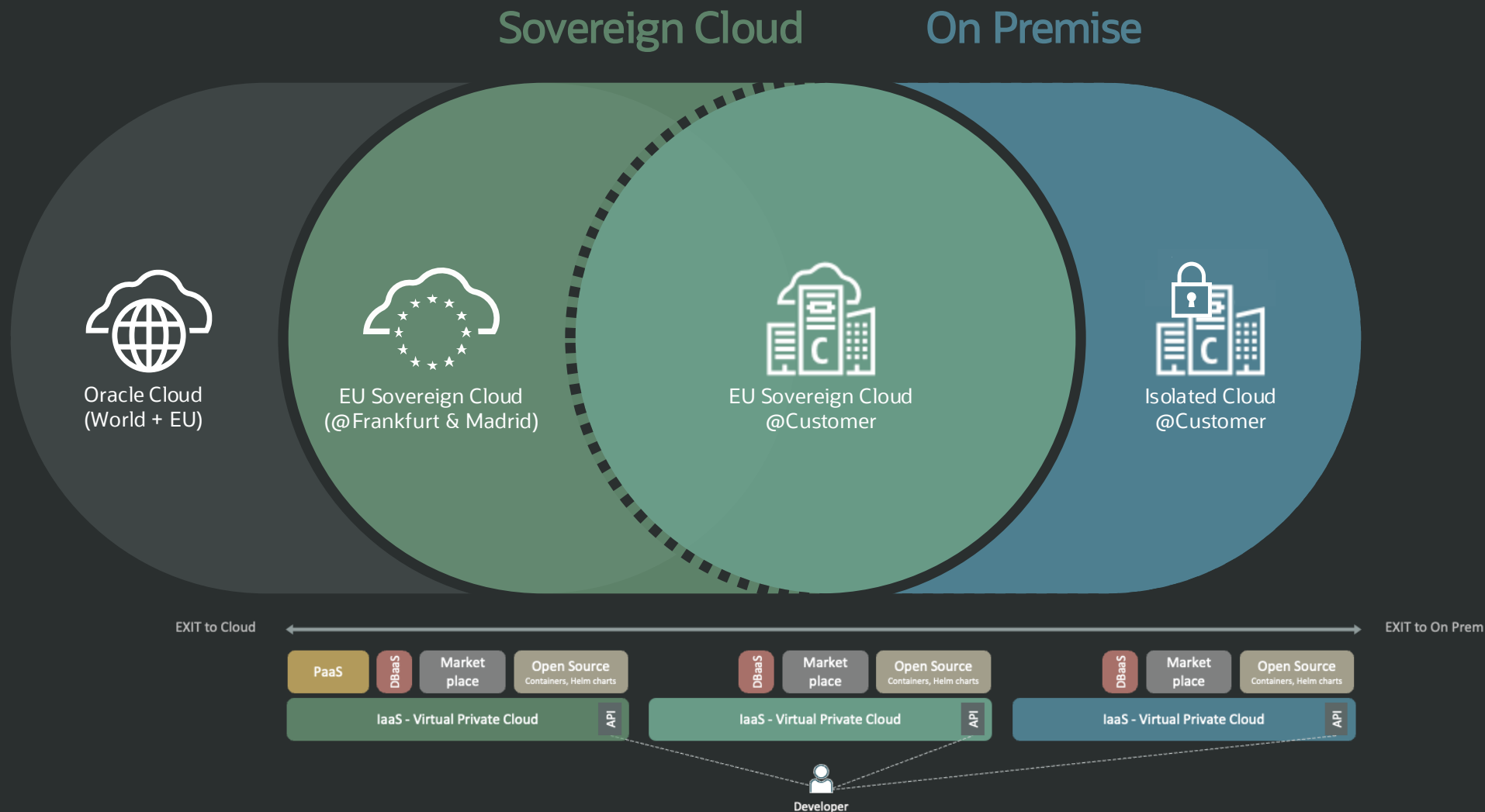
PHYSICAL ISOLATION FROM GLOBAL PUBLIC CLOUD



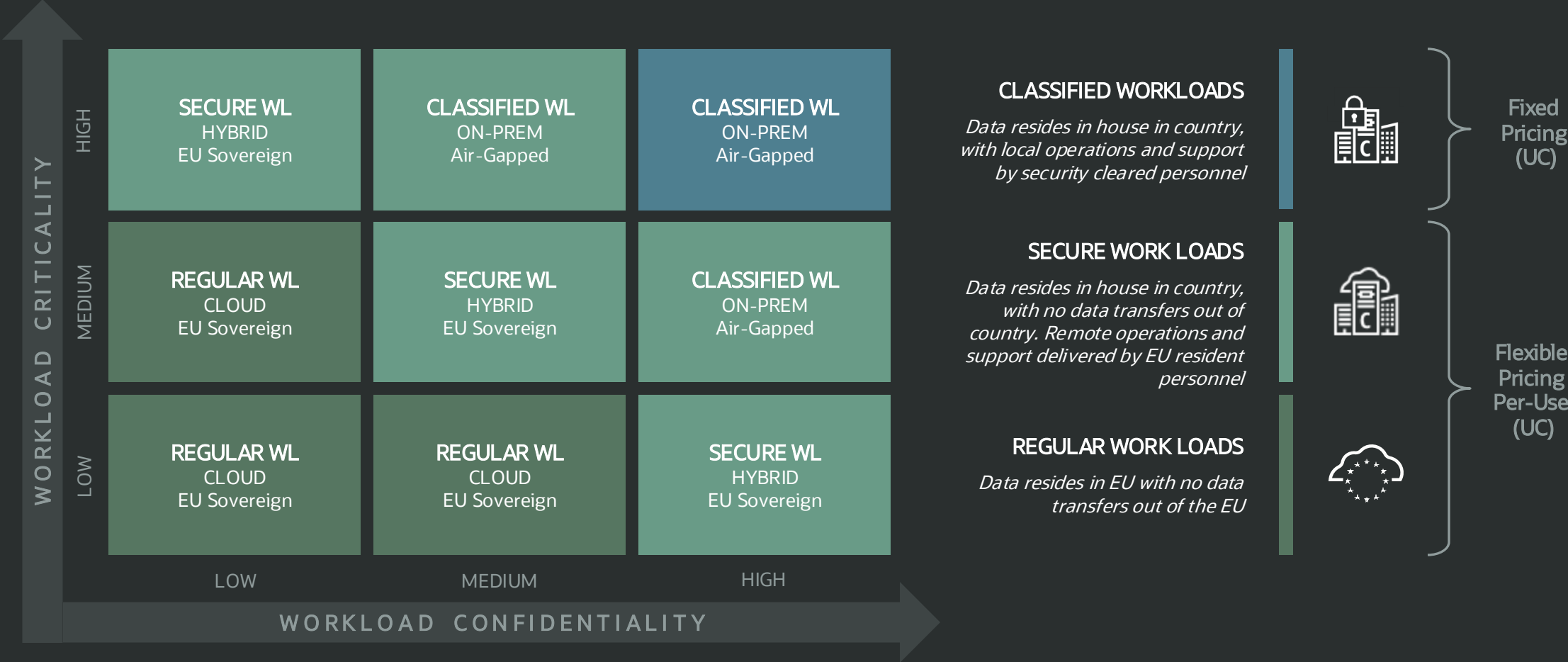
Oracle EU Sovereign Cloud

The first full Hyperscale Public Cloud
dedicated to the European Union

Sovereign and Hybrid Cloud – Combining the best from both worlds

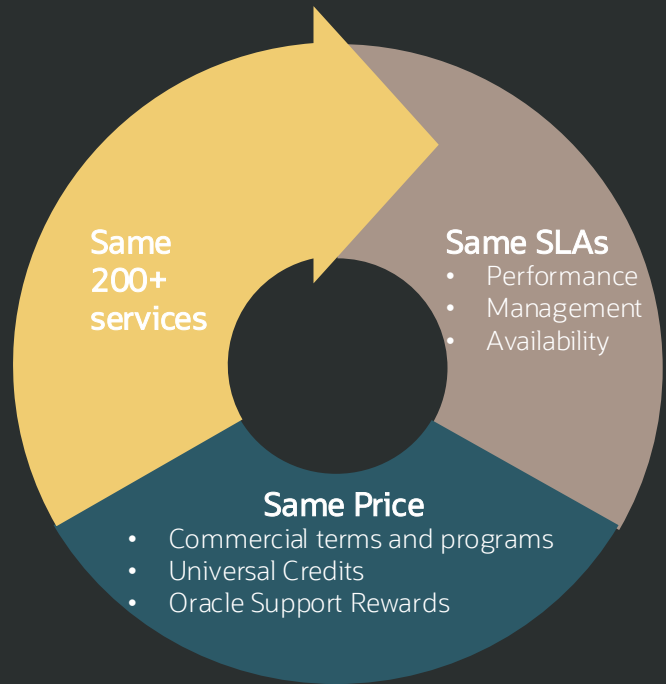


Sovereign and Hybrid Cloud – WL Classification and Distribution



Looking Inside of EU Sovereign Cloud

No trade-off between digital sovereignty and cloud capabilities



- Dedicated set of **organizational**, **technical** and **contractual** controls, with EU Sovereign Legal entities.
- **No joint venture or partnership complexities** in time to market, operations, features set, or depth of support.
- Customers benefit from the **same enterprise-grade support**, entirely located in the EU, at **no extra cost**.



Physically Isolated
EU Regions
Isolated EU
Network
Backbone



100% Asset
ownership by EU
Legal Entities



Compliance with
all EU and global
Cybersecurity
regulations



No Data Transfer
outside of the EU
territory

Organizational Controls

—
Overview

EU Sovereign Cloud Entities, Operations and Support

Complete separation from commercial regions



Operated by new, EU based legal entities created for this EU Sovereign region purpose.

These entities own:

Hardware and assets

Data Center leases

Cloud operations

Customer Support



The isolation of the EU Sovereign Cloud realm allows Oracle to restrict support and operations personnel to EU residents employed by the EU legal entities, including physical and logical access to the realm



The EU-based legal entities are backed by a governance committee to ensure integrity and alignment with current and future regulations

Support and operations are delivered within EU borders by EU residents employed by EU legal entities isolated from Oracle global cloud operations

EU Sovereign Cloud Dedicated Legal Entities

Oracle Sovereign Cloud
Germany GmbH

Oracle Sovereign Cloud
Spain, SRL

Responsible for providing support, hosting, and maintenance services for Oracle Sovereign Cloud Entities and for procuring EU Sovereign Cloud data centers

EU Sovereign South
(Madrid)

EU Sovereign Central
(Frankfurt)

Oracle Sovereign Cloud
Ireland Ltd

Oracle Sovereign Cloud
Polska Sp.z.o.o.

Oracle Sovereign Cloud
Czech Republic S.R.O

Oracle Sovereign Cloud
Romania S.R.L

Oracle Sovereign Cloud
Tech **Romania** S.R.L

Responsible for providing support, hosting, and maintenance services for Oracle Sovereign Cloud entities



COMPLETE EU
OPERATIONS

Technical Controls

Overview

EU Sovereign Cloud helps users control data at multiple levels

Oracle offers unique proven technical features based on real-world experiences



Isolated EU Sovereign Cloud Realm

- Dedicated EU Sovereign Cloud Realm provides a full physical and logical separation from any other Oracle Cloud Regions
- Realm access is only possible to EU Sovereign legal entities teams: deployment, operation, support, maintenance, security
- This is providing customers with by design and by default EU data localization and EU data residency



Security and access controls, data protection and governance

- Full stack of cybersecurity solutions on top of EU Sovereign Cloud dedicated realm strong foundation
- Customers can enhance EU Sovereign Cloud deployments by implementing security controls to meet their audit, compliance and sovereignty goals
- Advanced data encryption and key management options for data security and confidentiality



Resiliency for applications and data

- Full Disaster Recovery capabilities available today entirely within the EU Sovereign Cloud realm
- High availability and resilience can be implemented locally, each Sovereign Region containing 3 Fault Domains
- FDs are separate groupings of hardware and infrastructure, they let customers distribute instances on different physical hardware within a Data Center

Advanced data encryption capabilities and control

Oracle offers unique and proven technical features based on real-world experiences



Data at rest and in transit encrypted at all times

- **Data at rest encrypted by default – encryption cannot be turned off**
 - Block volumes, Object storage buckets, File storage, Database services (Autonomous, ExaCS)
 - Encryption using AES-256
- Data in transit encrypted with TLS1.2 and Media Access Control security (MACSec) for point-to-point security on Ethernet links



Data in use can be encrypted through confidential computing

- Data in use protection through memory encryption with enhanced isolation using AMD Secure Encrypted Virtualization (SEV)
 - Data in memory isolated via encryption keys
 - **No code change required for onboarding**
 - Minimal performance impact across most workloads
 - **Confidential compute shapes same price as same non confidential shapes**
 - Bare Metal instances available



Advanced key storage and management

- **Complete Key Management choice:**
 - Oracle managed for simplicity
 - Customer managed for control
- Many Vault type: Software Vault, Shared or Private HSM partition
- Unique capabilities for advanced customers Sovereignty requirements:
 - Dedicated single-tenant HSM with no OCI APIs in Oracle EUSC
 - External Key Management: customers use On-Premise KMS/HSMs:
 - “Bring Your Own Key” (BYOK)
 - “Hold Your Own Key” (HYOK) with Thales KMS

Dedicated Region | What is it?



COMPLETE EU
OPERATIONS

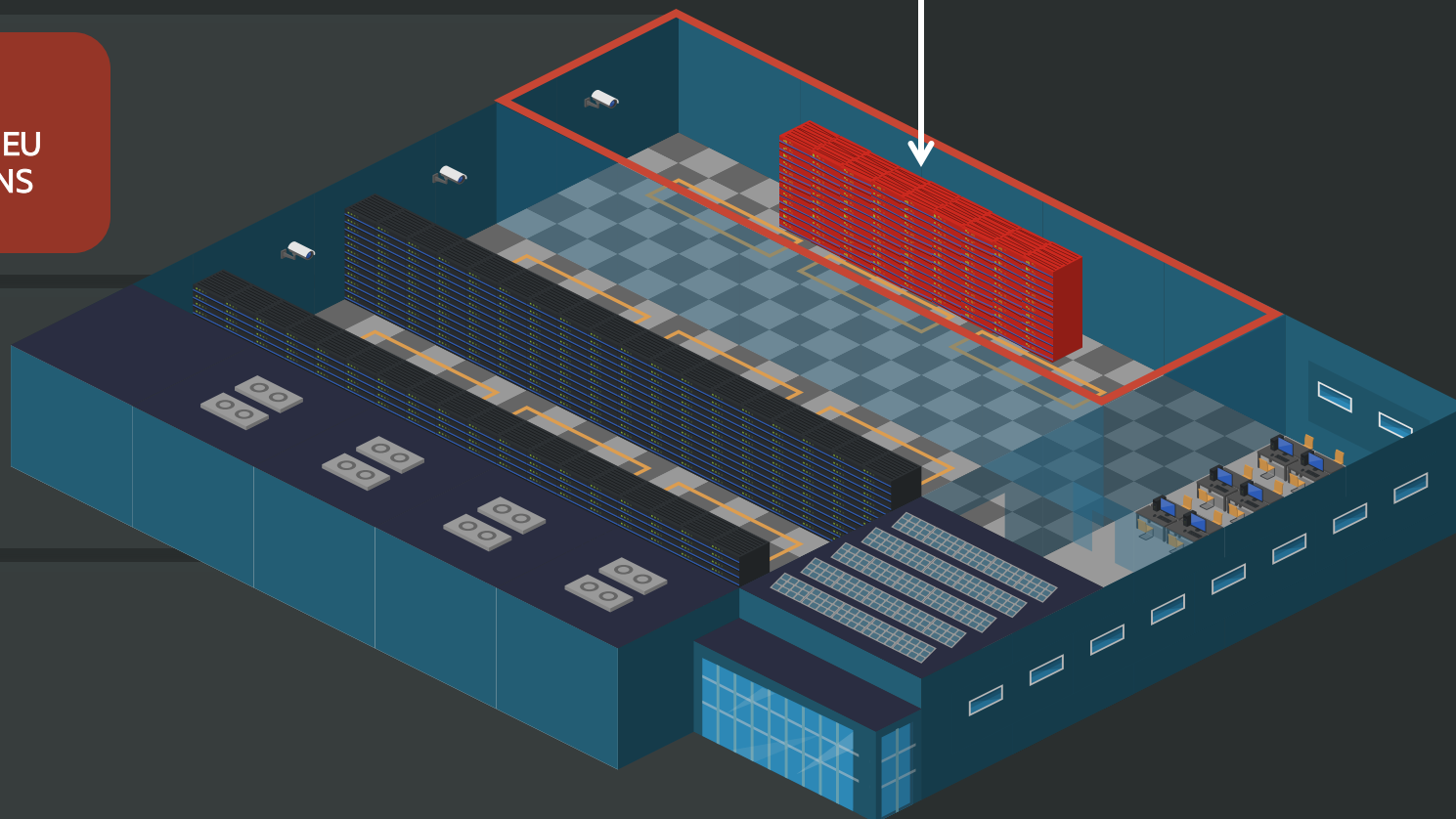


Full Cloud



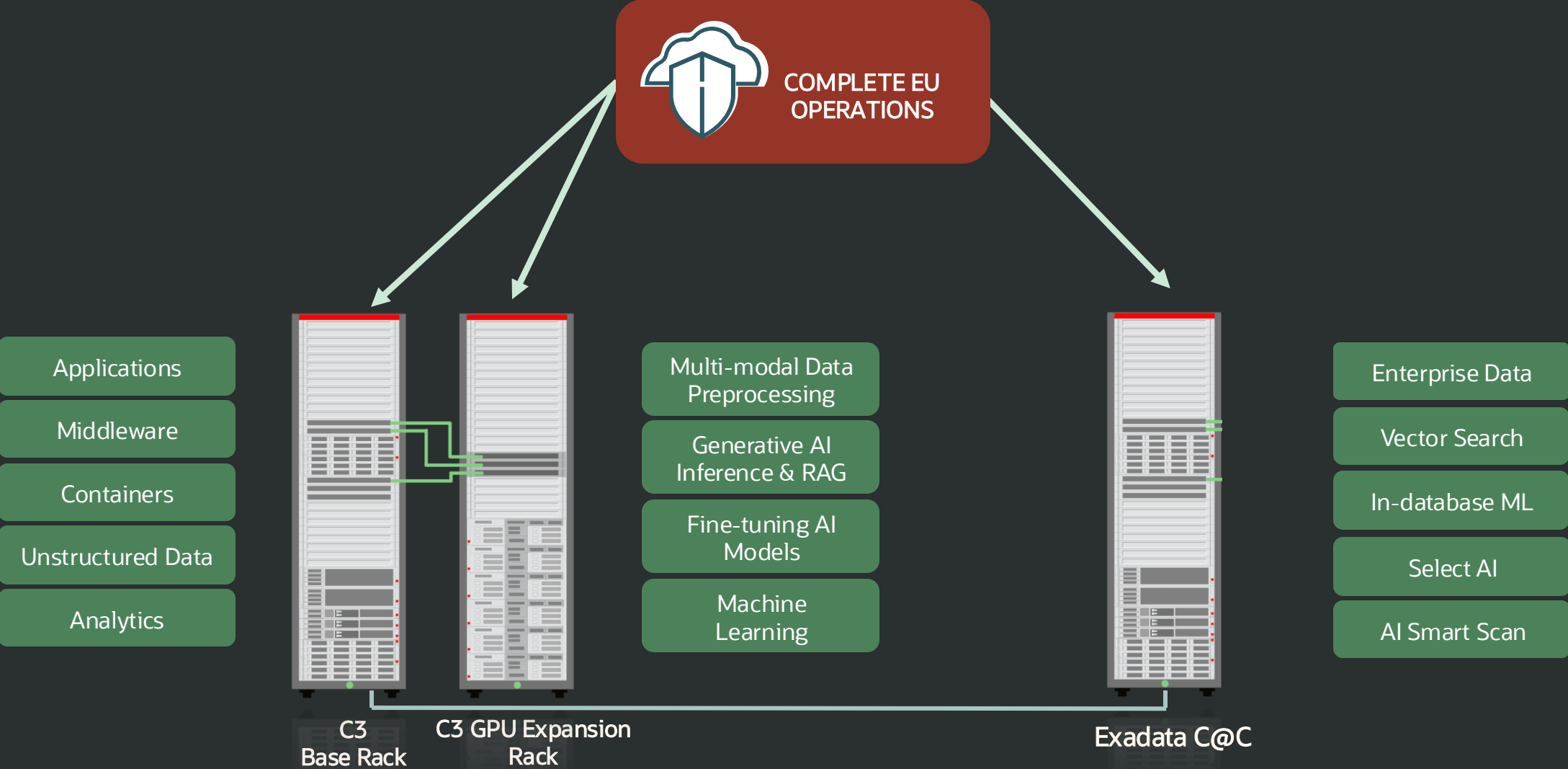
Your Cloud

ORACLE
CLOUD
Infrastructure



Customer data center

Full EU Sovereign Operations for Compute Cloud @ Customer



Contractual Controls

Overview

Contractual controls for EU Sovereign Cloud: why they matter



Supplement technical and organization controls to provide adequate level of data protection



Provide clarity and assurance for legal compliance



Define responsibilities between Oracle and EU Sovereign Cloud customers

“In principle, supplementary measures may have a contractual, technical or organisational nature. Combining diverse measures in a way that they support and build on each other may enhance the level of protection and may therefore contribute to reaching EU standards.”

Source: [EDPB Recommendations 01/2020](#) Link: [edpb_recommendations_202001vo.2.0_supplementarymeasurestransferstools_en.pdf](#)

EU Sovereign Cloud Can Uniquely Help Your Business



Your platform for critical workloads

Run and modernize the most challenging workloads with greater architectural flexibility and superior performance, all in a Sovereign Cloud within the EU



Superior cloud pricing and flexibility

Save money on your most critical cloud resources, EU Sovereign Cloud prices are the same as our low price Global Commercial Cloud OCI



The full cloud in the EU by EU employees only

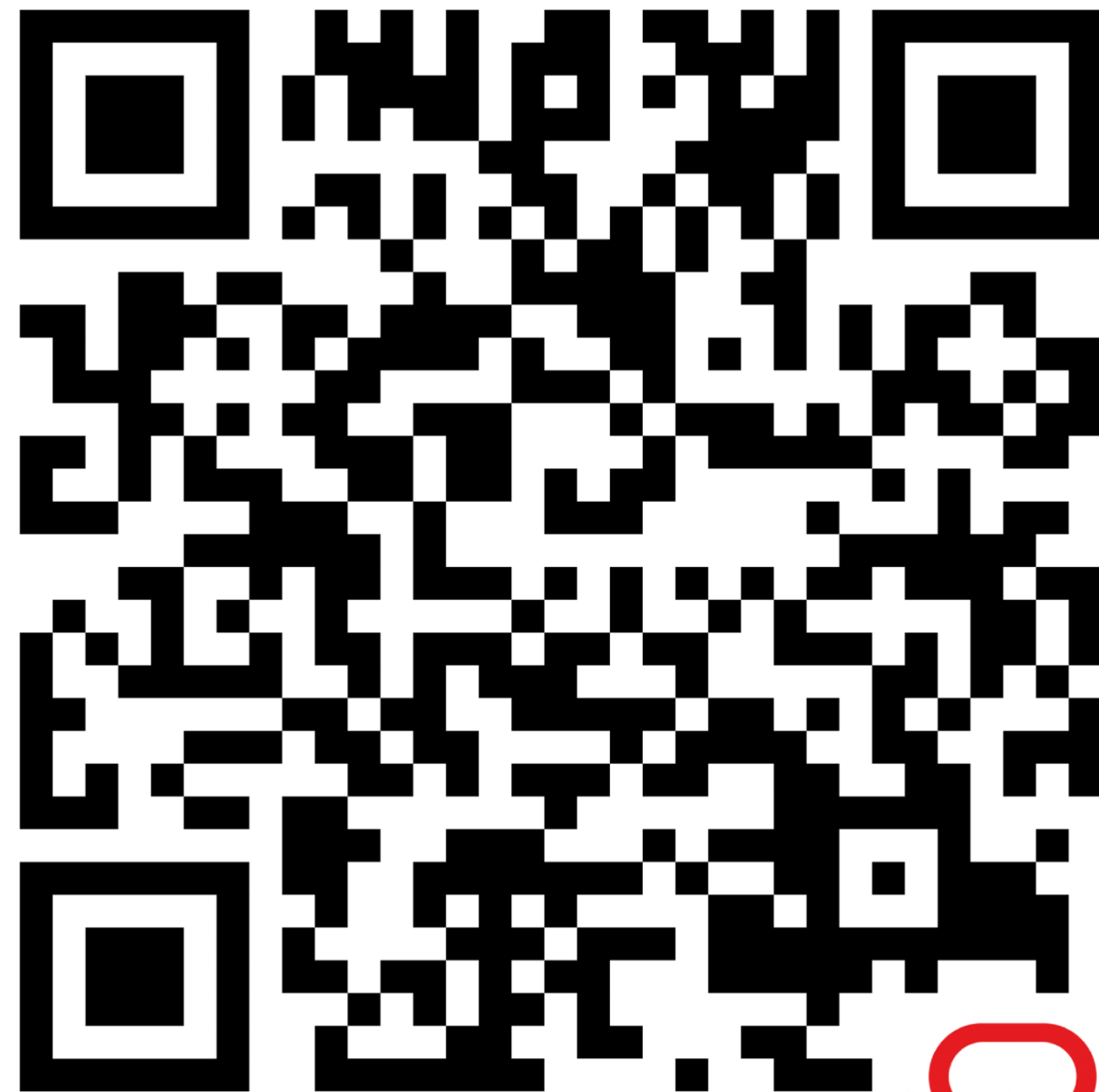
Use OCI Services in our EU Sovereign Cloud without compromising on Services, SLA, Price and Sovereignty



The path to Sovereign AI business value

AI for your business processes, your data and applications, and for builders – with no transfer of your data outside the EU

Q&A





Thank you!